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Does Social Media Dominate Government Report Cards in Influencing Nursing Home Demand?

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Abstract. *Problem definition:* Social media has become an indispensable platform for disseminating quality information to consumers across various service sectors. Recently, it has extended its influence to healthcare services, which traditionally relied on government report cards to disclose standardized quality information to the public. This article explores the impact of social media on consumer demand for healthcare services and compares its effectiveness with government report cards. *Methodology/results:* We analyze quality ratings of U.S. nursing homes collected from two information channels: (1) consumer ratings on Yelp and (2) government ratings on Nursing Home Compare, both of which adopt a five-star quality rating scale and are accessible on the Internet. We employ the method of difference-in-differences with continuous treatment intensity and instrumental variables to analyze the data. Using nursing home resident admissions as a proxy for consumer demand, we find that higher Yelp ratings led to higher consumer demand, particularly among Medicare-covered consumers. Furthermore, the effect of Yelp ratings was primarily driven by extreme ratings (one-star or five-star), as opposed to neutral ratings. We also find that Yelp ratings exerted a stronger effect on consumer demand than government ratings. This dominance of Yelp ratings over government ratings was observed primarily in markets with high Yelp penetration or markets with low and medium consumer education levels. Although higher Yelp ratings were associated with increased net incomes, we find little evidence that nursing homes made quality improvement in response to their Yelp ratings. *Managerial implications:* We recommend that the Centers for Medicare and Medicaid Services recognize social media platforms as valuable sources of information and collaborate with reputable platforms, such as Yelp, to promote public awareness of government report cards like Nursing Home Compare. Moreover, we advise nursing home operators to proactively manage their reputation on social media by promptly addressing consumer complaints and implementing quality improvement measures.

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Keywords: healthcare • social media • service operations • online consumer reviews • government report cards • nursing homes • quality of care • textual analysis

1. Introduction

Social media has become an indispensable platform for disseminating quality information to consumers across various service sectors, ranging from hotels to restaurants to home repairs. By reducing the information asymmetry of quality, social media has helped consumers make better decisions in choosing service providers. Recently, social media has extended its influence to healthcare services, providing consumer ratings and reviews for various healthcare providers such as hospitals, nursing homes, and physicians. Some social media platforms specialize in one type of healthcare service. For example, RateMDs and Zocdoc provide consumer ratings and

reviews for physicians only. Other social media sites such as Yelp and Google have added multiple healthcare services to their existing consumer review platforms.

Social media is not the only channel for consumers to gather quality information about healthcare services. In the United States, the Centers for Medicare and Medicaid Services (CMS), a federal agency within the Department of Health and Human Services, has long been disclosing healthcare providers' quality information to the public through online report card systems such as Hospital Compare and Nursing Home Compare.¹ These online government report cards provide quality ratings of individual healthcare providers based on inspections

conducted by government agencies and/or on self-reported data. The academic literature has shown that quality ratings from government report cards are strongly associated with patient health outcomes (Cornell et al. 2019) and induce consumers to choose healthcare providers with higher ratings (Perraillon et al. 2019).

In contrast to these standardized government report cards, quality information on social media is contributed by individual consumers and thus might be subjective and biased. Indeed, we find that healthcare providers tend to receive either the highest or the lowest consumer ratings based on our data gathered from Yelp. Furthermore, because healthcare services are generally infrequently used, most healthcare providers have not been rated or reviewed, and the number of ratings or reviews per provider is small. Because of these deficiencies, critics question the credibility and effectiveness of social media as a quality information channel for healthcare services. This concern has motivated our research questions: Given the presence of government report cards, does social media affect the consumer demand of healthcare services? If so, which quality information channel has a stronger effect on consumer demand?

In this study, we seek to answer these questions in the context of U.S. nursing homes by exploring longitudinal consumer rating data from Yelp (www.yelp.com) and government rating data from Nursing Home Compare (www.medicare.gov/care-compare/?guidedSearch=NursingHome) between 2010 and 2018. We choose Yelp to study the effect of social media because it is one of the largest online consumer review platforms in the United States and has included nursing homes on its platform since 2005. On Yelp.com, consumers can review nursing homes and rate them using a five-star rating scale. Yelp computes a *composite rating* of a nursing home by taking the average of its historical consumer ratings. We will refer to this composite rating as the Yelp rating of a nursing home to distinguish it from individual consumer ratings. Nursing Home Compare is a CMS-maintained online report card system to publicly disclose quality information of all Medicare- and Medicaid-certified nursing homes in the United States. In December 2008, CMS enhanced Nursing Home Compare by launching the Five-Star Quality Rating System (FSQRS) to provide an easy-to-understand overall quality rating (CMS 2019). Such a unique setting allows us to directly compare the impact of Yelp ratings and government ratings on consumer demand; that is, we can compare the effects of a one-star increase in Yelp ratings and a one-star increase in government ratings.

As the first step of our empirical analysis, we apply a difference-in-differences (DID) model with continuous treatment intensity (Duflo 2001) to estimate the effect of Yelp ratings on resident admissions of nursing homes. Because we do not observe a nursing home's demand directly, we use its total resident admissions

and Medicare-covered resident admissions as the proxy for its consumer demand. The latter measure more accurately reflects consumer preference of nursing homes because Medicare-covered residents generally do not experience nursing bed shortages and take priority over other types of residents such as Medicaid-covered residents. To deal with the endogeneity of Yelp ratings, we use reviewers' ratings of other businesses on Yelp, for example, restaurants and hotels, to construct an instrumental variable (IV) for Yelp ratings of nursing homes. The underlying assumption of this IV is that a reviewer's preference is stable over time across different businesses. And a reviewer's average rating of other businesses is naturally correlated with her rating of a nursing home due to a similar personal rating preference (e.g., a harsh reviewer tends to give low ratings to all businesses while a lenient reviewer tends to give high ratings), but it does not directly affect resident admissions of the nursing home. Using this IV, we show that resident admissions were higher in nursing homes with higher Yelp ratings. On average, a one-star increase in the Yelp rating of a nursing home led to a 2.5% increase in its total resident admissions and a 4.7% increase in the admissions of Medicare-covered residents.

To decompose the effect of Yelp ratings by the number of individual consumer ratings, we show that an additional one-star (five-star) individual rating decreases (increases) a nursing home's total and Medicare admissions, whereas those intermediate individual ratings do not have a significant impact on admissions. These results are consistent with the findings from our textual analysis of nursing home reviews—Using natural language processing (NLP) tools, we show that the sentiment scores of consumer reviews are positively correlated with their ratings and the effect of Yelp ratings is mostly associated with extreme reviews (with positive or negative sentiment).

We next compare the effects of Yelp ratings and government ratings on consumer demand. To do this, we re-estimate the aforementioned DID regression by adding one-year lagged government ratings as an IV for government ratings, thereby treating both Yelp and government ratings as explanatory variables. We find that a one-star increase in its Yelp rating increased a nursing home's Medicare admissions by 2.6% more than a one-star increase in its government rating did.

To understand why Yelp ratings exert a stronger effect on consumer demand than government ratings, we conduct two heterogeneous analyses related to brand recognition and consumer literacy. We show that the dominance of Yelp ratings over government ratings in influencing nursing home demand was only present in markets with high Yelp penetration or markets with low and medium consumer education levels. These heterogeneous results suggest that Yelp's brand recognition for other businesses, for example, restaurants, helps to

increase consumer awareness of its nursing home ratings. And consumers with lower education level might be less likely to be aware of government ratings or less receptive to government ratings because of the difficulty in digesting the relatively more complex quality information disclosed by government report cards.

Finally, we demonstrate the effect of Yelp ratings on nursing homes' financial performance. We find that a one-star increase in a nursing home's Yelp rating would increase its revenue by about 1.7% and its net income by about \$33,454 on average. Although these results suggest that it might be financially beneficial for nursing homes to improve their Yelp ratings, we find little evidence that nursing homes made quality improvement in response to their Yelp ratings.

To the best of our knowledge, our paper is one of the first to systematically compare the effects of social media and government report cards on consumer demand of services. These two quality information channels coexist in a range of service settings that are of high public interest, for example, hospitals, schools, restaurants, etc. In the U.S. healthcare industry, government report cards have long existed to disseminate quality information to the public, whereas social media has only emerged recently as an alternative channel of quality information. Our study suggests that consumer ratings on Yelp exert a stronger effect in influencing nursing home demand than government ratings on Nursing Home Compare. This finding is particularly surprising given that Yelp ratings are limited in quantity and only cover roughly a quarter of U.S. nursing homes, whereas Nursing Home Compare provides objective quality information for all U.S. nursing homes. Hence, our study calls for the attention of healthcare policymakers and government agencies to increase public education and consumer awareness of Nursing Home Compare, which CMS has devoted considerable resources to establish and maintain. To nursing home operators, we recommend that they recognize the significant influence of social media on consumer demand and consider improving quality accordingly.

2. Literature Review

Our work contributes to the online word-of-mouth (WOM) literature. This literature consists of two major streams of studies. One stream focuses on the generation of online reviews by studying which products are more likely to receive online reviews, what consumer characteristics lead to a positive or negative review, and how to filter fake reviews (Dellarocas and Wood 2008, Luca and Zervas 2016, Hu et al. 2017). The other stream studies how online reviews affect consumers' purchase decisions and firms' behaviors. The prevailing contexts studied in this stream include film reviews on Yahoo! Movies (Duan et al. 2008, Chintagunta et al.

2010), book reviews on Amazon and Barnes & Noble (Li and Hitt 2008, Sun 2012), and restaurant reviews on Yelp (Luca 2016).

All the aforementioned studies consider the effect of online reviews on experience goods, whose quality consumers have sufficient knowledge to evaluate. Nevertheless, there is a lack of understanding about how online reviews affect consumer demand of credence goods, whose quality cannot be well observed even after consumption, and typical examples include expert services such as healthcare services and automobile repairs.

Recently, a small but growing stream of works study the effect of online reviews on credence goods in the context of physician services. For example, Gao et al. (2015) show that physicians with lower quality are less likely to be rated online due to the effect of "sound of silence"—patients usually refrain from complaining about their dissatisfied experience because they believe it may impact their future treatment. In contrast, Lu and Rui (2018) find that physicians with no rating information deliver better patient outcomes than those with low ratings and deliver no worse outcomes than those with high ratings. Saifee et al. (2019) analyze physician ratings on [Vitals.com](https://www.vitals.com) and find a positive correlation between ratings and the use of electronic health record (EHR). Luca and Vats (2013) and Xu et al. (2021) show that physicians with high ratings are associated with high demand. Chen (2018) analyzes physician ratings on Yelp and concludes that a physician with a high rating would receive more patient volume, resulting in increased revenue. To our surprise, the impact of online reviews and ratings on consumer demand of healthcare facilities, such as hospitals and nursing homes, is underexplored thus far.

More broadly, our work contributes to the literature on the use of information technology in operations. This is a rapidly growing research field, and here we only selectively review a few papers that are more closely related to our study. A common theme of this literature is to examine whether information technology improves operational efficiency and reduces misinformation in a variety of industries including agriculture (Parker et al. 2016), retailing (Liu et al. 2021, Calvo et al. 2023), sharing economy (Cui et al. 2020), labor hiring (Belavina et al. 2020), and healthcare services (Li et al. 2022, Delana et al. 2023). Different from these studies, we study how information disclosure on social media affects consumer demand of healthcare services.

Our work also contributes to the literature of government report cards, which exist in a range of service settings with high public interest, e.g., education, restaurants, and healthcare services (Jin and Leslie 2003, Hurwitz and Smith 2018, Kim and Martin 2021). A large body of research exists for quality disclosure of nursing homes (Lu 2016, Werner et al. 2016, Perrillon et al. 2019). Most

of these studies use empirical analysis to show that quality disclosure through government report cards affects healthcare providers' quality. Recently, Saghaian and Hopp (2020) develop a stylized choice model to study the patients' choices of hospitals in response to government report cards. They find that social welfare is improved after quality disclosure, but such an improvement diminishes in the long term.

Although both the online WOM literature and the government report cards literature have been well developed in their own ways, the relationship between consumer ratings and government ratings is not clear. Are consumer ratings of credence goods substitutes or complements for government ratings? Do consumers have sufficient medical knowledge to evaluate the quality of healthcare services? Recently, some researchers took an initial step in studying the relationship between the two channels of quality information. Bardach et al. (2013) find that Yelp and government ratings for hospitals are positively correlated. Johari et al. (2018) compare Yelp and government ratings for nursing homes in California. They find that there is a positive, but weak correlation between Yelp and government ratings and the former is lower than the latter. Ranard et al. (2016) and Kellogg et al. (2018) qualitatively analyze patterns of consumer reviews on Yelp and find that more than half of the reviews are related to staff attitudes and responsiveness which government report cards do not cover. In contrast to these studies, our study empirically estimates *consumer responses* to Yelp ratings and government ratings using a nationwide U.S. nursing home data set. To the best of our knowledge, our work is the first to systematically study the difference between the effects of social media and government report cards on consumer demand of healthcare services.

3. Institutional Background and Hypothesis Development

3.1. U.S. Nursing Home Residents

3.1.1. Types of Nursing Home Residents and Their Care Needs. A nursing home, also known as a skilled nursing facility, is a place for people who do not need to be in a hospital but can no longer be cared for at home, according to the definition of the National Institute on Aging. There are three types of nursing home residents categorized by their payer sources: Medicare, Medicaid, and private paying. Nursing home care of the first two types is paid by the federal and state governments, whereas that of the last type is either paid by private long-term care insurers or out of residents' own pockets. In 2016, Medicare covered 14% of U.S. nursing home residents, whereas Medicaid covered 62% (Kaiser Family Foundation 2018).

Medicare-covered residents generally get admitted into nursing homes for *short-term postacute* care after

they are discharged from hospitals where they have undergone a major surgery or medical treatment. Medicare is a national health insurance program for people who have worked and paid taxes for at least 10 years. These taxpayers and their spouses are qualified for Medicare when their age reaches 65 or above. It covers up to 100 days of nursing home care after a medically necessary inpatient hospital stay (CMS 2018). The average length of stay (LOS) of Medicare residents in a nursing home is around 39 days (MedPAC 2018, p. 224) and the average daily reimbursement rate is about \$526.

Medicaid-covered residents are mostly the older elderly, generally aged 85 and above, with chronic diseases. These residents are economically disadvantaged because their eligibility for Medicaid coverage requires their income and assets to be below state-imposed limits. Different from Medicare residents, Medicaid residents require *long-term* care with an average LOS being around two years (Jiang et al. 2020). Medicaid's daily reimbursement rate is about \$230, much lower than Medicare's.

3.1.2. Process of Choosing a Nursing Home. Medicare-covered residents are transferred to nursing homes directly after they are discharged from hospitals, and thus typically choose nursing homes before being discharged. Physicians, physical therapists, and social workers jointly suggest whether a patient should be discharged home or to a nursing home (Jin et al. 2022a). Specifically, a physician decides whether a patient is ready to be discharged from the hospital. Next, a physical therapist suggests a discharge destination such as a nursing home, home, or other places. If discharge to a nursing home is recommended, a social worker then provides a list of candidate nursing homes. The patient and her family make the final call on the specific discharge destination. Patients take into account costs, quality, and facility availability when determining their discharge destinations. Medicare allows patients to pick any Medicare-certified nursing home, which accounts for 85% of nursing homes in the U.S. Medicare generally covers the total cost for the first 20 days, whereas residents pay about \$186 per day during days 20–100 and pay all the costs after 100 days. Medicaid-covered residents generally choose nursing homes close to their original community based on the quality and availability of facilities. They and their family members often visit facilities to get first-hand information.

Quality is one of the most important factors in a resident's nursing home choice. The elderly and their family members usually rely on both online and offline channels to obtain quality information. The online rating information usually comes from social media (e.g., Yelp) or government report cards (e.g., FSQRS). The offline quality information could be obtained through word of mouth or visits to different nursing homes. Different

people might use different quality information channels depending on their awareness and personal preference.

Finally, the availability of a nursing home bed is a significant factor in an elderly's nursing home choice. Evidence suggests that serving short-stay Medicare residents can generate a higher profit margin than serving long-stay Medicaid residents, and therefore nursing homes generally prefer and compete for Medicare residents (Thomas 2015, Lu et al. 2021). Hence, Medicare residents have a great chance to get access to their preferred nursing homes. In contrast, Medicaid residents are often placed on a waiting list of their preferred nursing homes, and they might experience long waiting or have to settle for nursing homes other than their preferred. In other words, Medicare residents have much more freedom than Medicaid residents in choosing a nursing home.

3.2. Quality Information Channels

3.2.1. Social Media: Yelp. Yelp, one of the largest consumer review platforms in the United States, included nursing homes on its platform in 2005. Yelp is visited by 30 million unique devices each quarter and has accumulated more than 220 million reviews in its database by 2020.² Consumers can search for nursing homes by specifying the state and city. Yelp then returns a list of nursing homes within the location and displays general information such as facility name, address, phone number, and the average rating for each nursing home. Once a consumer clicks on a nursing home, Yelp displays more detailed information such as reviewers' names, ratings, comments, and review dates (see a screenshot in Figure A1(a) in the online appendix). Consumers can click on a reviewer's name to see her homepage, which displays photos, the total number of reviews, rating distribution, and historical reviews of all businesses she rated on Yelp.

3.2.2. Government Report Cards: Nursing Home Compare. To make nursing homes' quality information readily available to the public, CMS established Nursing Home Compare (NHC), an online report card system in 2002. Later, to make the quality information disclosed on NHC more understandable to the public, CMS launched the FSQRS in December 2008. This government rating system gives each nursing home a rating between one to five stars for each of the three quality dimensions: health inspections, staffing, and various quality of care measures (see a screenshot in Figure A1(b) in the online appendix). FSQRS also assigns each nursing home an overall star rating based on its aggregate performance in the three dimensions.

Health inspections of nursing homes are conducted by state inspection teams. A nursing home is randomly inspected every 12–15 months. This random inspection policy helps to mitigate the possibility of nursing home

manipulations. The inspection team checks quality along more than 190 dimensions mandated by CMS. If the quality of a certain dimension does not meet the minimum quality standard set by CMS, a deficiency citation will be issued for this quality dimension. CMS provides a chart to convert deficiency citations into quality scores (CMS 2018). Staffing levels are reported by nursing homes to CMS. It includes eight staffing measures and covers three types of nurses.

Other quality information collected by state inspection teams consists of various long-stay and short-stay resident physical and clinical measures. For example, it is important for the elderly constrained on beds to be turned from time to time, which helps to prevent pressure ulcers from growing on their backs. Hence, CMS introduced a quantifiable measure, the percentage of skilled nursing facility (SNF) residents with pressure ulcers that are new or worsened, to indicate whether a nursing home conducts frequent turns for bedbound residents. For another example, older adults are more likely to stay healthy if they can conduct daily activities such as dressing, bathing, and toileting by themselves, which helps to prevent their physical capabilities from degenerating quickly. However, it takes more staffing for a nursing home to allow the elderly to take care of themselves. To encourage this good practice, CMS introduced the activities of daily living (ADL) index, and the percentage of long-stay residents whose need for help with daily activities has increased to check whether a nursing home gives the elderly enough opportunities to conduct daily activities without the help of nurses.

3.3. Hypothesis Development

In this section, we develop a series of testable hypotheses to study the effect of Yelp ratings on nursing home demand and investigate the difference between the effects of Yelp ratings and government ratings.

3.3.1. Effect of Yelp Ratings on Nursing Home Demand. The existing online WOM literature has shown that consumer ratings and reviews can influence consumer demand of experience goods because they disseminate valuable quality information to consumers. This has been empirically shown in various contexts ranging from movies to books to restaurants (Chintagunta et al. 2010, Sun 2012, Luca 2016). In the context of credence goods, especially healthcare services, there have been heated debates on to what extent consumer ratings and reviews reflect service quality and whether consumers should trust them. Although patients or their families without sufficient medical training cannot accurately judge the quality of healthcare services, it is still possible for them to obtain useful quality information based on their own experiences. For example, family members of an elderly at least know whether she is

rehospitalized after staying in a nursing home, or they can observe whether her health condition is worse off (e.g., suffering from chronic bedsores). As shown in Figure A1(a) in the online appendix, a reviewer gave one star to a nursing home as she was unsatisfied with nurse responsiveness, cleanliness, dietary, and so on, based on her grandma's experience. In light of this, we believe that online consumer ratings for nursing homes convey valuable first-hand quality information to consumers. A rational consumer, in turn, would choose those nursing homes with high ratings. Therefore, we expect the following hypothesis to hold.

Hypothesis 1a. *Higher Yelp ratings lead to higher total resident admissions.*

As described in Section 3.1, U.S. nursing homes' main customers are Medicare-covered and Medicaid-covered residents. In the United States, consumer access to nursing home care is determined by licensed beds of nursing homes. Each nursing home is licensed to have a fixed number of beds in three categories: Medicare-dedicated, Medicaid-dedicated, and flexible beds. The flexible bed can be freely allocated between Medicare and Medicaid residents, whereas the dedicated beds can only be used by the residents with the designated insurance. Jiang et al. (2020) find that more than 96% of nursing homes are licensed to serve both Medicare and Medicaid residents. Among them, 17% have Medicare-dedicated beds, whereas only 2.4% have Medicaid-dedicated beds. The higher presence of Medicare-dedicated beds than Medicaid-dedicated beds among U.S. nursing homes implies that Medicare-covered consumers would have more access to nursing home beds. Moreover, researchers have shown that nursing homes compete for Medicare-covered residents due to their higher margins (Thomas 2015, Lu et al. 2021). This implies that nursing homes would give priority to Medicare-covered consumers when filling their beds.

Because of their high profit margins and dedicated beds available in nursing homes, Medicare-covered consumers generally enjoy immediate access to nursing home beds. By contrast, Medicaid-covered consumers face rationed admission even when there are Medicare-dedicated beds available. In areas where nursing homes have high occupancy rates, Medicaid residents might end up not being able to choose their most preferred nursing homes due to capacity constraints. Research has shown that about 20% of Medicaid-covered elderly were rationed out due to nursing homes' capacity constraints (Ching et al. 2015). Therefore, we conjecture that the effect of Yelp ratings on nursing home demand is reflected mostly in the admissions of Medicare residents, as stated in the following hypothesis. In other words, Medicare admissions is a more sensitive proxy for nursing home demand than total admissions.

Hypothesis 1b. *Higher Yelp ratings lead to higher Medicare resident admissions.*

3.3.2. Comparing the Effects of Yelp Ratings and Government Ratings. Although Yelp has been demonstrated to influence consumer demand of restaurants (Luca 2016), critics question the credibility of Yelp as a reliable quality information channel and its influence on nursing home demand for several reasons. First, different from other types of services reviewed on Yelp, the number of consumer ratings for each nursing home is very limited—it is only 5.3 per rated nursing home by 2018. Hence, the quality information conveyed on Yelp might be highly skewed by a few consumer ratings. As a result, the effect of fraudulent reviews on a facility's overall rating would be more pronounced for a nursing home than for a restaurant or a hotel. Nevertheless, Yelp has taken action to filter out those fraudulent reviews.³ Since 2015, Yelp has started showing evidence of fraudulent reviews directly to their users with screenshots and notes.⁴ Second, nursing home ratings on Yelp tend to be highly polarized—consumers are more likely to give either one-star or five-star ratings to a nursing home than to give medium ratings. Third, a large number (72%) of nursing homes have not received any consumer ratings. Consumers may interpret the quality of these unrated nursing homes in different ways according to some studies. Gao et al. (2015) show that patients may not give their physicians reviews after experiencing poor-quality services to avoid being labeled as “bad mouth.” Following this logic, rational consumers might view those unrated nursing homes as low-quality providers. It is therefore unclear what effect Yelp ratings may exert on nursing home demand.

As explained in Section 3.2.2, government ratings on NHC are constructed based on multidimensional quality performance measures, which not only are standardized but also provide guidance to nursing homes on what regulators care about (Brickley et al. 2021). Such government report cards have been shown to significantly increase quality information transparency and motivate firms to improve quality (Dranove and Jin 2010), and high government ratings are positively associated with future quality of care (Kimball et al. 2018, Cornell et al. 2019). Compared with Yelp ratings, the quality information disclosed by government report cards tends to be more credible than consumer ratings. Therefore, one would expect government ratings to exert a stronger influence on nursing home demand than Yelp ratings, as stated in the next hypothesis.

Hypothesis 2a. *Government ratings dominate Yelp ratings in affecting resident admissions.*

However, government report cards are not without shortcomings. In fact, government report cards have been shown to increase the risks that healthcare providers

game with government-led quality inspections. For example, Dranove et al. (2003) show that physicians tend to shun sicker patients when the government audits the mortality rate and discloses it to the public. Lu (2012) shows that nursing homes may take the teaching-to-the-test strategy to game with the government's quality inspection. The recent New York Times study shows that CMS five-star ratings are subject to the manipulating behaviors of nursing homes, making the ratings inconsistent with the true quality of rated facilities (Silver-Greenberg and Gebeloff 2021). In contrast, social media platforms allow consumers to evaluate any quality dimension they observe and provide quality information that can be easily digested by peer consumers. Therefore, social media platforms such as Yelp may be more credible from the perspective of consumers. For example, when a nursing home mistreated a resident, her family members might complain about such behavior on Yelp. In light of this, we expect the following hypothesis to hold.

Hypothesis 2b. *Yelp ratings dominate government ratings in affecting resident admissions.*

Yelp has established its reputation as a trustworthy online review platform for various consumer services. Therefore, Yelp's brand recognition helps to increase consumer awareness and the credibility of its nursing home ratings. In contrast, consumer awareness of government ratings is fairly low (Konetzka and Perraiillon 2016) and their impact on consumer demand is small (Werner et al. 2012). If Yelp ratings indeed dominate government ratings in influencing nursing home demand, we conjecture that the dominance of Yelp ratings over government ratings would be stronger in markets where Yelp has higher penetration.

Hypothesis 3. *The dominance of Yelp ratings over government ratings is stronger in markets with higher Yelp penetration.*

Furthermore, we conjecture that in markets with lower consumer education level, the dominance of Yelp ratings over government ratings might be stronger. There are three reasons for this conjecture. First, consumers with lower education level might be less likely to be aware of government ratings. Second, consumers with lower education level might have more difficulty processing various quality measures available on NHC and thus are less receptive to government ratings than those with higher education level (Jin et al. 2022b). Third, well-educated consumers may be more skeptical about Yelp ratings given a limited number of consumer ratings and the existence of fake reviews. By contrast, Jin et al. (2022b) show that consumers with low education level are insufficiently skeptical about non-disclosed information. Therefore, they might be more receptive to

Yelp ratings. Taken together, these arguments lead to the following hypothesis.

Hypothesis 4. *The dominance of Yelp ratings over government ratings is stronger in markets with lower education level.*

4. Data

4.1. Data Sources and Sample Construction

Our data set consists of three main data sources, the 2010–2018 Online Survey Certificate and Reporting Database (OSCAR), the Skilled Nursing Facility Cost Reports, and consumer ratings on Yelp. The OSCAR database is maintained by CMS and covers all Medicare- and Medicaid-certified nursing homes in the United States. It provides facility-level information including government ratings, detailed quality information (such as deficiency citations), staffing, and beds. The SNF Cost Reports provide detailed annual financial and admission information of individual nursing homes. These two data sets can be merged via the nursing home provider number. Yelp, a well-known online consumer review platform, provides a nursing home's name, address, phone number, and consumer ratings over time. We merge the Yelp data with the other two data sets using a nursing home's name, location, and phone number. In addition to these three data sets, we obtain resident characteristics from the Long-Term Care Focus data and market characteristics from the U.S. Bureau of Census.

We exclude all hospital-based skilled nursing homes⁵ from our main sample because reviews of a hospital-based nursing home usually contain reviews of other hospital departments, which might lead to a biased rating of the skilled nursing department. We also exclude all nursing homes with occupancy rates above the 95th percentile. Their resident admissions are not affected by their ratings because increased demand cannot be served due to capacity constraints.⁶ Our final sample consists of 13,099 skilled nursing facilities, of which 3,741 (28.6%) have been rated on Yelp by 2018 with 19,819 individual consumer ratings in total. Table 1 provides the definition and summary statistics of key variables used in our main sample.

4.2. Explanatory Variables: Nursing Home Quality Ratings

4.2.1. Consumer Ratings on Yelp. Yelp, one of the largest social media platforms in the United States, collects and displays consumer-contributed reviews and ratings of nursing homes. An individual consumer can voluntarily rate a nursing home on a scale of one (lowest) to five (highest) and provide textual reviews. We collected all nursing home ratings and reviews on Yelp up to 2018. Our data collection method is described in detail in Online Appendix A.

Table 1. Summary Statistics

Variable	Definition	Observations	Mean	Standard deviation
Yelp and government ratings				
Rated	1 if a nursing home was rated and 0 otherwise	117,891	0.10	0.30
YelpRating	Cumulative average individual ratings	11,541	2.84	1.45
OtherRating	Average of reviewers' average ratings to other businesses	11,541	3.47	0.93
NumRatings	Cumulative number of reviews	11,541	3.26	4.58
GovRating	Government ratings	117,891	3.11	1.35
Resident admission measures				
Total admissions	Admissions of total patients	117,891	279	223
Medicare admissions	Admissions of Medicare patients	117,891	141	133
Financial performance measures				
Net revenue	Total revenue deducted by contractual allowances and discounts on patients accounts	117,891	8,807,351	5,359,657
Net income	After-tax net income (dollars in thousands in regression)	117,891	93,839	933,928
Quality measures				
Total citations	Total counts of deficiency citations	104,792	6.12	5.11
30-day rehospitalization	Percentage of patients who were readmitted to hospital from the hospital within 30 days	104,792	18.26	6.94
Control variables				
RN staffing level	RN nurse hours per resident day	117,891	0.55	0.34
Occupancy	Occupancy rate	117,891	0.81	0.15
Percentage Black	Percentage of Black patients	117,891	10.85	18.57
Percentage female	Percentage of female patients	117,891	67.64	11.76
Average age	Average age of patients	117,891	80.00	6.12
ADL index	ADL index describing patient health status	117,891	16.45	2.60
HHI	Competition measure at the county level	117,891	0.23	0.25
Log income	Log of income per capita at the county level	117,891	10.67	0.26
Log population	Log of population at the county level	117,891	12.21	1.77

A nursing home's overall Yelp rating is displayed on the platform and is generated using the mean of all past consumer ratings.⁷ We do not directly observe historical Yelp ratings but can reconstruct a nursing home's yearly Yelp rating by calculating its cumulative average individual consumer ratings up to the end of that year, which we will refer to as "Yelp rating" to distinguish it from individual consumer ratings. Therefore, our measure of Yelp ratings only mimics what consumers observed at any time in the past. We will address this measurement error using an instrumental variable strategy (Section 4.2).

In our Yelp sample, nursing home ratings were contributed by 18,765 individual consumers, around 4.4% of whom have rated more than one nursing home. Figure 1(a) shows the total number of consumer ratings increased from 302 in 2010 to 19,819 in 2018, a 65-fold increase. Figure 1(b) shows that the average number of

consumer ratings per rated nursing home is increasing steadily over time, reaching 5.3 by 2018. Figure 1(b) also shows that the earlier a nursing home got its first rating, the more consumer ratings a nursing home would get over time.

In the online WOM literature, the distribution of individual consumer ratings is usually positively skewed, asymmetric, and bimodal, which is known as J-shaped (Hu et al. 2017, Lu and Rui 2018). The J-shaped consumer rating distribution is explained by the theory of "sound of silence" in the context of online physician ratings—consumers with bad experiences of physicians tend to keep silent to avoid being labeled as "bad mouth" (Gao et al. 2015). Consumer ratings of all businesses on Yelp have this J-shaped distribution. In contrast, consumer ratings of nursing homes on Yelp have a U-shaped distribution (Figure 1(c))—reviewers were more likely to give extreme ratings, that is, one star or five stars, to nursing

Figure 1. (Color online) Distribution of Nursing Home Ratings on Yelp



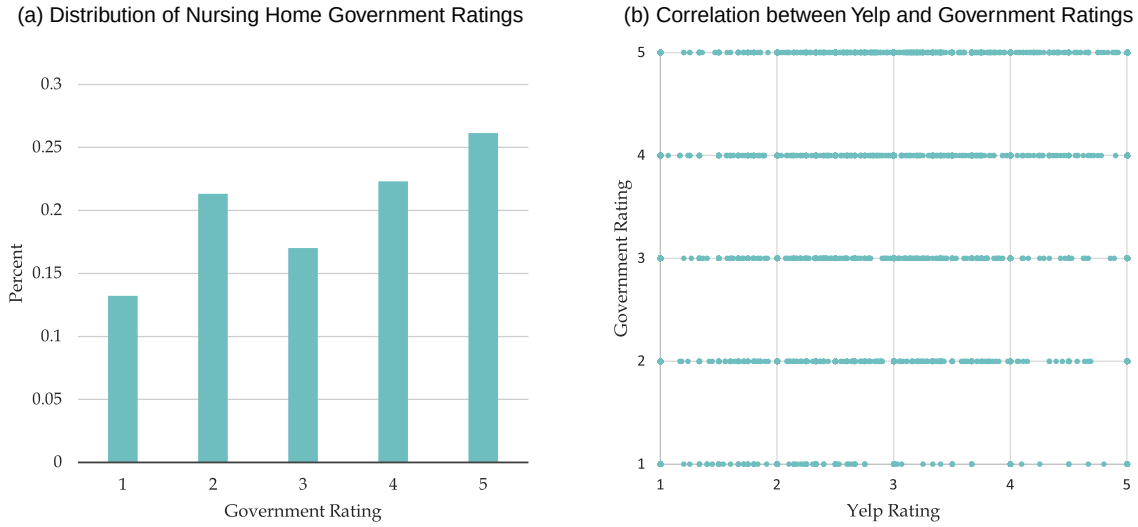
homes. One explanation for the U-shaped distribution is that reviewers might be more likely to contribute reviews online after having extremely good or bad experiences (Li and Hitt 2008). In contrast to Figure 1(c), which shows the distribution of individual consumer ratings, Figure 1(d) shows the distribution of nursing homes' Yelp ratings, which has a mean of 2.71 in 2018. Note that a nursing home's Yelp rating is a composite rating, as defined earlier, and thus is a continuous measure (up to 0.5).

4.2.2. Government Ratings. As described in Section 2.2.1, NHC's FSQRS provides an overall rating to each nursing home based on its aggregate performance in three quality domains. We use these overall ratings in our subsequent analysis and refer to them as government ratings.

The mean of government ratings is 3.27 and the distribution of government ratings is displayed in Figure 2(a).

Analyzing the relationship between Yelp ratings and government ratings reveals that the correlation between them is fairly weak—the correlation coefficient is 0.20 (Figure 2(b)). This observation is consistent with the finding of Johari et al. (2018) that Yelp ratings differ considerably from government ratings for nursing homes in California.

4.3. Dependent Variables: Resident Admissions
One of the key dependent variables of our analysis is resident admissions. Most of the existing online WOM literature uses sales revenue as a proxy for consumer demand assuming uniform pricing across consumers

Figure 2. (Color online) Distribution of Nursing Home Government Ratings

(Jin and Leslie 2003, Dai and Luca 2020). In our setting, nursing homes' revenue per resident differs across resident types due to their different payer sources. Therefore, instead of using revenue, we use resident admissions as a proxy for consumer demand. Specifically, we use both total admissions and Medicare admissions as dependent variables. Total admissions include the admissions of all residents, whereas Medicare admissions include only the admissions of Medicare-covered residents. Recall that Medicare admissions better reflects consumer preference of nursing homes as we explained in Section 3.3.1.

5. Empirical Methods

5.1. Panel Data Estimation

To study the effect of Yelp ratings on nursing home demand, we apply a DID model with continuous treatment intensity (Duflo 2001). The unit of observation in our analysis is a nursing home in a specific year, and our baseline model is specified as

$$Y_{it} = \alpha_0 + \alpha_1 \text{Rated}_{i,t-1} + \alpha_2 \text{YelpRating}_{i,t-1} \times \text{Rated}_{i,t-1} + \alpha_3 \text{GovRating}_{i,t-1} + \alpha_4 \text{NumRatings}_{i,t-1} + \alpha_5 X_{it} + \alpha_6 Z_{it} + \theta_i + \mu_t + \epsilon_{it}. \quad (1)$$

In Equation (1), the dependent variable, Y_{it} , represents nursing home i 's resident admissions in log transformation in year t . There are two key explanatory variables in the base model. First, $\text{Rated}_{i,t-1}$ is a binary variable that equals one if nursing home i has been rated by the end of year $t-1$ and equals zero otherwise. Second, $\text{YelpRating}_{i,t-1}$ represents nursing home i 's Yelp rating (i.e., cumulative average rating) by the end of year $t-1$. For both Rated and YelpRating , we use one-year lagged variables for two purposes. First, they mimic what consumers

observe online in year t . Second, it is likely there exists a delay between when consumers observe a nursing home's rating and when they make their nursing home decisions. Among the coefficients in Equation (1), α_1 captures the effect of being rated by Yelp, whereas α_2 captures the differential effect of Yelp ratings conditional on being rated. Hence, $\alpha_1 + n\alpha_2$ is the total effect of getting an n -star Yelp rating, where $n \in \{1, \dots, 5\}$.

In the baseline model, we include several control variables. For example, $\text{GovRating}_{i,t-1}$ represents nursing home i 's government rating in year $t-1$. We include this one-year lagged government rating to control for its effect on nursing homes' resident admissions. $\text{NumRatings}_{i,t-1}$ represents the cumulative number of consumer ratings nursing home i has received by year $t-1$. We include this control variable because consumers are more likely to trust a Yelp rating when it is generated from a larger number of consumer ratings (Duan et al. 2008); X_{it} is a vector of nursing home characteristics such as the staffing level of registered nurses (RNs) and resident characteristics such as the ADL index; and Z_{it} is a vector of market characteristics at the county level, including competition intensity measured by the Herfindahl-Hirschman Index (HHI), log of income per capita, and log of population. In addition, we include nursing home i 's fixed effect (θ_i) to control for any time-invariant unobserved factors affecting the nursing home and year fixed effect (μ_t) to control for unobserved factors affecting all nursing homes in year t . ϵ_{it} is the error term. We cluster standard errors by nursing homes.

5.2. Identification and Instrumental Variable Strategy

There are two potential endogeneity issues in the baseline model specified by Equation (1). First, there might

exist unobserved time-varying factors that affect both a nursing home's Yelp rating and its resident admissions. For example, nursing homes may have already been improving their facility amenities in response to their past Yelp ratings and thus are more likely to receive higher ratings from consumers. This would likely attract more consumers to nursing homes at the same time. For another example, consumers' offline activities such as direct visits to local nursing homes would presumably affect both Yelp ratings and nursing home demand. These are omitted variable issues.

Second, we construct a nursing home's Yelp rating in year $t - 1$ by computing the cumulative average rating by the end of year $t - 1$. However, what consumers actually observed before they chose a nursing home in year t was the cumulative average rating up to the time when they visited Yelp's website. If additional consumer ratings were contributed during year t but before consumers visited Yelp's website, then the Yelp ratings used in our analysis might be slightly different from what the consumers observed in year t . This is a measurement error issue.

To correct for these potential endogeneity issues, we use an IV approach to establish the causal relationship between Yelp ratings and resident admissions. Specifically, we construct a time-varying IV for Yelp ratings. Specifically, we use reviewers' ratings of other businesses on Yelp as IVs for their ratings of nursing homes. This approach has been adopted by Chen (2018) in his study of physician ratings. To be explicit, we define n_{it} as the cumulative number of reviewers for nursing home i by the end of year t . For each reviewer $k \in K(n_{it}) = \{1, 2, \dots, n_{it}\}$, who has rated nursing home i by year t , we define r_{-i}^k as reviewer k 's average rating of all other businesses, i.e., all businesses except nursing home i . Hence, we define the IV for $YelpRating_{i,t}$ as nursing home i 's cumulative average of its reviewers' ratings of other businesses by year t :

$$IV \text{ for } YelpRating_{i,t} \triangleq OtherRating_{i,t} = \frac{1}{n_{it}} \sum_{k \in K(n_{it})} r_{-i}^k. \quad (2)$$

From now on, we will refer to this IV as the *OtherRating* IV. The intuition behind this IV is that a reviewer's average rating of other businesses reflects her personal rating preference, that is, whether she is a lenient or harsh reviewer. Hence, it is highly correlated with her rating of a nursing home. However, her average rating of other businesses would not directly affect the resident admissions of the nursing home she rated. This is because a reviewer's ratings of other businesses on Yelp are generally irrelevant to a consumer's offline activities such as chatting with friends and family and direct visits to local nursing homes that might affect resident admissions. Therefore, this IV allows us to address the omitted

variable issue caused by those unobserved offline activities. Further notice that each reviewer's average rating of other businesses has equal weight in the calculation of $OtherRating_{i,t}$ so that a frequent reviewer's rating would not dominate an infrequent reviewer's rating.

In the case that nursing home i 's reviewers did not rate any other businesses, we set r_{-i}^k as their ratings of nursing home i . This definition is imperfect but does not affect our results—we demonstrate the robustness of the results by excluding these reviewers' nursing home ratings or setting the IV of these reviewers' ratings to be equal to the average of other reviewers' ratings of other businesses (see Section 6.3.1 for details).

When running the two-stage least square regression (2SLS), we regress $YelpRating_{i,t-1} \times Rated_{i,t-1}$ against $OtherRating_{i,t-1} \times Rated_{i,t-1}$ in the first stage, and Table A1 in the online appendix shows the result. The coefficient of $OtherRating_{i,t-1} \times Rated_{i,t-1}$ is 0.68, positive, and at a 1% significance level. Moreover, the Kleibergen-Paap Wald rk F statistics can reject the hypothesis of weak IV. This suggests that reviewers' ratings of other businesses are valid IVs for Yelp ratings of nursing homes.

5.3. Selection Bias of Being Rated on Yelp

Rated nursing homes on Yelp might not be randomly distributed. In the existing literature of online WOM, researchers often consider the arrival of consumer ratings as random and assume that the sample of rated products or services is representative of the full sample (Gu et al. 2012). Nevertheless, we address the endogeneity issue that might be caused by the selection bias as follows.

The nursing home fixed effect (θ_i) in the base model (Equation (1)) has already controlled for any time-invariant factors (e.g., nursing home size) that might cause systematic differences between rated and unrated nursing homes. However, being rated might be correlated with some unobserved time-varying factors. To address this concern, we conduct a test developed by Semykina and Wooldridge (2010), which is an extension of the classic Heckman's selection model. The test consists of two steps. First, we conduct a probit analysis to derive the inverse Mills ratios (IMRs) using a binary dependent variable, $Rated_{ia,t-1}$, which equals one if nursing home i has been rated in area a (a county or a hospital referral region) by the end of year $t - 1$ and equals zero otherwise. We construct an exogenous explanatory variable using Yelp ratings of hospitals located in the same area as nursing home i . This variable is correlated with the probability of nursing home i being rated on Yelp but does not directly affect its resident admissions. In other words, we use Yelp penetration for hospital ratings as an instrument for $Rated_{ia,t-1}$. Second, we include the IMRs in the second stage and test the significance

of its coefficient. The details of this test are described in Online Appendix B, and the results are reported in Table A2 in the online appendix. This table shows that the coefficient of IMR is not statistically significant, which implies that the time-varying selection bias in our sample is negligible and thus unlikely to cause a bias of our estimator in the base model.

Nevertheless, we conduct a robustness check using propensity score matching to address any potential selection issue (see Section 6.3.3 for details). We shall point out that even if rated nursing homes in our sample were not randomly distributed, they contribute to 28.6% of all U.S. nursing homes, already a significant representation, thereby making them important to study in their own right.

6. Main Results

6.1. Effect of Yelp Ratings on Nursing Home Demand

We first examine the effect of Yelp ratings on nursing homes' resident admissions. Table 2 reports the results on total admissions and Medicare admissions. The first two columns report the ordinary least squares (OLS) results, and the second two columns report the 2SLS results using the *OtherRating* IV. These results are both qualitatively and quantitatively similar.

Let us focus on the 2SLS results. Column (3) of Table 2 shows the effect of Yelp ratings on total admissions. The coefficient of *Rated* is -0.05 , indicating a baseline negative effect of being rated. The coefficient of $\text{YelpRating} \times \text{Rated}$ is 0.025 , indicating that a one-star increase in the Yelp rating led to a 2.5% increase in a nursing home's total

resident admissions conditional on being rated. Both coefficients are statistically significant. Taken together, they imply that compared with those not being rated on Yelp, a five-star Yelp rating increases total admissions by 7.6%, whereas a one-star Yelp rating has a statistically insignificant negative effect on total admissions.⁸ These results support Hypothesis 1a.

Moreover, the effect of Yelp ratings on Medicare admissions is stronger than on total admissions, as shown in column (4). The coefficient of *Rated* is -0.110 , indicating a baseline negative effect of being rated. The coefficient of $\text{YelpRating} \times \text{Rated}$ for Medicare admissions is 0.047 , indicating that a one-star increase in the Yelp rating led to a 4.7% increase in a nursing home's Medicare admissions conditional on being rated. Taken together, they imply that compared with those not being rated on Yelp, a five-star Yelp rating increases Medicare admissions by 12.5%, whereas a one-star Yelp rating decreases Medicare admissions by 6.2%. These results support Hypothesis 1b.

Next, we study the impact of rating distribution instead of ratings. Namely, we replace Yelp ratings in Equation (1) with the number of one-star individual consumer ratings (i.e., extremely low ratings), the number of five-star individual ratings (i.e., extremely high ratings), and the number of two-star, three-star, and four-star individual ratings (i.e., relatively neutral ratings). These measures allow us to decompose the effect of Yelp ratings over different categories of individual consumer ratings. The results are presented in Table A3 in the online appendix. Columns (1) and (2) shows that conditional on being rated on Yelp, an additional one-star (five-star) individual

Table 2. Effect of Yelp Ratings on Resident Admissions

	OLS		IV for <i>YelpRating</i> only		Both IVs	
	Log total admissions (1)	Log Medicare admissions (2)	Log total admissions (3)	Log Medicare admissions (4)	Log total admissions (5)	Log Medicare admissions (6)
<i>Rated</i>	-0.029^* (0.012)	-0.049^{**} (0.015)	-0.050^* (0.023)	-0.110^{***} (0.029)	-0.050^* (0.023)	-0.109^{***} (0.029)
$\text{YelpRating} \times \text{Rated}$	0.017^{***} (0.004)	0.025^{***} (0.005)	0.025^{**} (0.008)	0.047^{***} (0.010)	0.025^{**} (0.008)	0.047^{***} (0.010)
<i>GovRating</i>	0.009^{***} (0.001)	0.014^{***} (0.002)	0.009^{***} (0.001)	0.014^{***} (0.002)	0.015^{***} (0.003)	0.021^{***} (0.004)
1-star <i>YelpRating</i>	-0.012	-0.024^*	-0.025	-0.062^{**}	-0.025	-0.062^{**}
5-star <i>YelpRating</i>	0.058^{***}	0.075^{***}	0.076^{***}	0.127^{***}	0.074^{***}	0.126^{***}
F-test: $\text{YelpRating} \times \text{Rated} - \text{GovRating}$	0.009^*	0.010^*	0.016^*	0.033^{**}	0.010	0.026^*
Time-varying controls	Yes	Yes	Yes	Yes	Yes	Yes
Facility fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Year fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Observations	117,891	117,891	117,891	117,891	117,891	117,891
R^2	0.876	0.859	0.876	0.859	0.876	0.859

Notes. The table shows the estimation results of Equation (1) using the full nursing home sample from 2010 to 2018. All the regressions include nursing home fixed effects, year fixed effects and time-varying controls (RN staffing level, occupancy rates, %Black, %female, average age, ADL index, HHI, log of income per capita, and log of population). Standard errors in parentheses are clustered at the nursing home level.

⁺ $p < 0.1$; ^{*} $p < 0.05$; ^{**} $p < 0.01$; ^{***} $p < 0.001$.

rating decreases (increases) a nursing home's total and Medicare admissions, whereas those intermediate ratings do not have a significant impact on admissions.

To validate the effect of Yelp ratings, we conduct a textual analysis on nursing home reviews. The details of the analysis are relegated to Online Appendix C. Here we summarize and report the results. Specifically, we use the biterm topic model (Yan et al. 2013) to extract frequently mentioned topics, and use Valence Aware Dictionary and sEntiment Reasoner (VADER) (Hutto and Gilbert 2014) to identify the corresponding sentiment for each topic. This allows us to categorize the review texts into six categories and construct measures of topic-based sentiments for nursing home reviews. These six categories are clinical care, facility, safety, staffing, financial issues, and general issues (Table A4 in the online appendix). For each nursing home, we construct six topic-based sentiment scores for each review and then compute the cumulative average of its reviews' sentiment scores for each topic category. We use these sentiment scores to replace the composite Yelp ratings in Equation (1) and re-estimate the DID specification. The results are displayed in Table A5 in the online appendix. This table shows that the sentiment scores of three topic categories—clinical care, facility, and staffing—have a positive effect on Medicare admissions, whereas only those of facility and staffing have a positive effect on total admissions. These results suggest that review texts on Yelp do influence nursing home demand. The sentiment scores of these categories with significant effects also have fairly high correlations with Yelp ratings, as shown in Figure A2 in the online appendix.

Besides, we use VADER to identify the corresponding sentiment for each consumer's review. We then construct three variables, namely, the number of reviews with positive sentiment, the number of reviews with negative sentiment, and the number of reviews with neutral sentiment, and use them to replace the variable of Yelp ratings in Equation (1). The results in columns (3) and (4) of Table A3 in the online appendix show that an additional positive review increases resident admissions, whereas an additional negative review decreases resident admissions. Together with columns (1) and (2), these results suggest that resident admissions are mostly affected by the number of extreme ratings (one star or five stars) and extreme reviews (positive or negative sentiment) instead of those neutral ratings or reviews.

6.2. Comparing the Effects of Yelp Ratings and Government Ratings

In this section, we compare the effects of Yelp ratings and government ratings in influencing nursing home demand. In particular, we re-estimate Equation (1) using the *OtherRating* IV for Yelp ratings (see Section 5.2) and using lagged government ratings ($GovRating_{i,t-2}$)

as an IV for government ratings ($GovRating_{i,t-1}$). It is worth noting that throughout our entire sample period of 2010–2018, government ratings exist for all nursing homes because CMS introduced the FSQRS on NHC in December 2008.

Although Yelp ratings are the cumulative average consumer ratings, government ratings of nursing homes are constructed using the most recent quality inspection information. In addition, consumers can view all the historical consumer ratings for a nursing home on Yelp, but only observe the most recent government ratings on NHC. Therefore, we can safely assume that the government rating of a nursing home in year $t - 2$ is highly correlated with its government rating in year $t - 1$ but does not directly affect its resident admissions in year t . This approach of using the lagged variable as an instrument has been widely used in the management literature (Archak et al. 2011, Ghose et al. 2012).

Table 2, columns (5) and (6), reports the results using the lagged-variable IV for government ratings and the *OtherRating* IV for Yelp ratings. These two columns show that the coefficients of $YelpRating \times Rated$ are positive and significant for both total admissions and for Medicare admissions. The coefficients of $GovRating$ are also positive and significant for both total admissions and Medicare admissions, suggesting that higher government ratings would result in higher admissions.

To compare the effects of Yelp ratings and government ratings, we perform an F test on their coefficients ($YelpRating \times Rated - GovRating$). The test reveals that a one-star increase in Yelp ratings increased Medicare admissions by 2.6% more than a one-star increase in government ratings did, but no significant difference existed between their effects on total admissions. These observations support Hypothesis 2b but do not support Hypothesis 2a.

Interestingly, Yelp ratings and government ratings are usually inconsistent, and their correlation coefficient is only 0.20. In fact, a nursing home receiving a five-star government rating might be receiving a one-star Yelp rating (Figure 2(b) and Figure A1 in the online appendix). Two reasons might explain such inconsistencies. First, these two ratings focus on different quality dimensions, as we described in Section 3.2, and consumers' perceived quality might be different from quality measures of government inspections. Second, government ratings might be inaccurate—a recent article in the New York Times reported that CMS's five-star ratings of nursing homes might not be accurate because they are partly determined by self-reported data that are subject to nursing homes' manipulation (Silver-Greenberg and Gebeloff 2021).

Given that Yelp ratings and government ratings are not always consistent, how would they together affect consumer demand? To answer this question, we compute the joint effect of the two ratings using the coefficients

Table 3. Joint Effects of Yelp and Government Ratings on Total and Medicare Admissions

(a) Total Admissions						(b) Medicare Admissions					
Government Ratings	Yelp Ratings					Government Ratings	Yelp Ratings				
	1	2	3	4	5		1	2	3	4	5
	0.034	0.059***	0.084***	0.109***	0.134***		0.021	0.068**	0.115***	0.162***	0.209***
	0.020	0.044**	0.069***	0.094***	0.119***		0.000	0.047**	0.094***	0.141***	0.188***
	0.005	0.030*	0.054***	0.079***	0.104***		−0.021	0.026 ⁺	0.073***	0.120***	0.167***
	−0.010	0.015	0.040***	0.064***	0.089***		−0.041*	0.006	0.053***	0.100***	0.147***

Notes. The two matrices show the joint effects of Yelp and government ratings on Total (left) and Medicare (right) admissions, using the results from Table 2, columns (5) and (6). Each row represents a government star rating and each column refers to a Yelp star rating on a one to five scale. We calculate the joint effects using the formula $\text{coeff. of Rated} + \text{coeff. of YelpRating} \times \text{Rated} \times \text{Number of Yelp stars} + \text{coeff. of GovRating} \times (\text{Number of government stars} - 1)$.

⁺ $p < 0.1$; * $p < 0.05$; ** $p < 0.01$; *** $p < 0.001$.

from Table 2 and report the results in Table 3 in two 5×4 matrixes—The left matrix is for the joint effect on total admissions (panel A), whereas the right one is for Medicare admissions (panel B). Each row represents a government rating level from two to five, whereas each column represents a Yelp rating level from one to five. The numbers in the matrixes represent the proportional increase in total or Medicare admissions as a result of getting the respective Yelp and government ratings when compared with the benchmark of no Yelp rating and a one-star government rating. Observe that receiving a higher than three-star Yelp rating would increase a nursing home's total admissions and Medicare admissions even if it receives a below three-star government rating (see the last rows of panels A and B). Taken together, these results suggest that Yelp ratings exert a stronger effect on nursing home demand than government ratings.

6.3. Robustness Checks

In this section, we conduct a series of robustness checks to ensure our results are valid. For brevity, all tables for these robustness check results are relegated to the online appendix.

6.3.1. Exclusion Criterion of the *OtherRating* IV. There could exist some omitted variables that jointly affect nursing home demand and the probability of people reviewing other businesses in a region over time. To alleviate this concern, we add *Country* $\times$ *year* dummy or *State* $\times$ *year* dummy in Equation (1) to control for region-specific time trends that might affect both reviewers' ratings and nursing home demand. The results shown in Table A6 in the online appendix remain consistent with the baseline results in Table 2.

Recall that we use reviewers' ratings of other businesses on Yelp to construct the IV for nursing home ratings. In our sample, 11.7% of nursing home ratings are contributed by reviewers who did not rate any other businesses. In our baseline IV strategy, we set the r_{-i}^k to be equal to consumers' ratings of nursing home i when they did not rate any other businesses. One might worry that this approach could violate the exclusion criterion. To address this concern, we conduct two robustness checks.

First, for those reviewers who did not rate any other businesses, we use the mean of r_{-i}^k of other reviewers (i.e., the mean of other reviewers' ratings of other businesses) to construct the IV as specified by Equation (2). This imputed IV does not include any ratings of nursing homes and thus satisfies the exclusion criterion. The results shown in columns (1) and (2) of Table A7 in the online appendix are similar to those in Table 2. We also report robustness checks using alternative definitions of the mean of r_{-i}^k , which is averaged within the same county or the same hospital referral region (HRR). See columns (3)–(6) in Table A7 in the online appendix for details.

Second, we exclude the ratings contributed by the reviewers who did not rate any other businesses. We end up with 3,488 nursing homes being rated with 17,495 ratings in total. In the remaining subsample, all reviewers rated other businesses and their average ratings of other businesses are used as the IV for Yelp ratings. Columns (7) and (8) in Table A7 in the online appendix report the results, which are similar to those in Table 2.

6.3.2. Inclusion Criterion of the *OtherRating* IV. Another concern over the IV for Yelp ratings (i.e., the *OtherRating* IV) is that reviewers on Yelp might have heterogeneous preferences over different product and

service categories and thus the IV might not satisfy the inclusion criterion. To tackle this concern, we take the following two approaches.

First, we seek to understand the relationship between reviewers' ratings of nursing homes and various business categories. To do so, we utilize the Yelp business category list⁹ to identify the category of each facility or service that has been rated by reviewers who have rated nursing homes on Yelp. We focus on the top nine most reviewed categories on Yelp, that is, restaurant, local and home services, shopping, beauty and fitness, health, auto, travel and hotel, arts and entertainment, and nightlife.¹⁰ For each reviewer, we calculate her average ratings of businesses within the same category. These average ratings reflect her differential rating preferences for various categories, for example, whether she is a lenient reviewer for restaurants or a harsh reviewer for shopping malls. Analyzing the relationship between a reviewer's nursing home rating and her average ratings for each category reveals that the variation among these correlations is small (Figure A3 in the online appendix). The category of health gives the highest correlation with nursing home ratings (0.33), whereas the category of nightlife gives the lowest correlation with nursing home ratings (0.20). The similarity in correlations suggests that consumers may exhibit similar behaviors when rating different categories.

Second, we construct an alternative IV using ratings of a subset of other businesses instead of all other businesses. Specifically, we reconstruct r_{-i}^k using reviewers' ratings of restaurants, home services, shopping, and health. We choose these four categories because reviewers' ratings of these categories have the highest correlation with Yelp nursing home ratings. In the case that a consumer did not rate any business within these four categories, we replace it with the average of r_{-i}^k of reviewers who have rated at least one business within these four categories. We use alternative definitions of the mean of r_{-i}^k , which is averaged within the same county or the same HRR. The results are shown in Table A8 in the online appendix, which are similar to those in Table 2.

6.3.3. Measurement Error of Ratings and Other Robustness Checks

Measurement Error of Ratings. In our base model, we use the one-year lagged Yelp and government ratings as our explanatory variables, which might be different from what consumers observed when they chose a nursing home during year t . To address this concern, we replace both ratings by the end of year $t - 1$ with those average ratings during year t . The results are shown in Table A9 in the online appendix, and they are consistent with the baseline results in Table 2.

Falsification Test. Critics may argue that Yelp ratings might be highly noisy given the limited number of

consumer ratings per nursing home. To ensure that changes in resident admissions are indeed driven by Yelp ratings rather than our model specification, we conduct a falsification test in which we randomly generate a fake Yelp rating for each rated nursing home each year using a uniform distribution with the support $[1, 5]$ and regress resident admissions on the fake Yelp ratings and other control variables following a specification similar to Equation (1). We simulate fake Yelp ratings 50 times and report the sample mean of coefficients and the corresponding confidence intervals in Table A10 in the online appendix. The falsification test shows insignificant coefficients for both *Rated* and *YelpRating* $\times$ *Rated*, which confirms that randomized fake ratings do not affect resident admissions and thus supports the validity of our results.

Propensity Score Matching. We use a propensity score matching (PSM) method to find appropriate controls for each facility that has been rated on Yelp. To do this, we first define a control pool including all nursing homes that were not rated on Yelp during the entire sample period. We denote the dependent variable to equal one if a nursing home was rated and zero otherwise. The explanatory variables include both resident and facility characteristics and county demographics. Because nursing homes were rated in different years, we run a logit model on the sample constructed from each individual year to compute the propensity score. After running the matching procedure, we assemble a sample of 56,592 observations over the sample period. We use this matched sample for the analysis and present the result in Table A11 in the online appendix. From this table, we observe a similar effect of Yelp ratings as that of the baseline result.

7. Mechanism Discussion

We have shown that Yelp ratings exert a stronger effect on nursing home demand than government ratings. In this section, we discuss two potential mechanisms and provide evidence to support the conjectures.

7.1. Brand Recognition: Yelp's Market Penetration

One possible explanation for the stronger effect of Yelp ratings is that Yelp is a popular online rating platform for consumer services and has higher brand recognition among consumers than NHC, which provides government ratings of nursing homes. In contrast, government ratings for nursing homes are not highly recognized by the public (Konetzka and Perrillon 2016), and their impact on consumer choice is fairly small (Werner et al. 2012). If so, we conjecture that the dominance of Yelp ratings over government ratings would be stronger in markets where Yelp has higher penetration, as we stated in Hypothesis 3 in Section 3.3.2.

To proceed, we use hospital ratings on Yelp to construct a measure of Yelp's market penetration. We

Table 4. Heterogeneous Analysis: Yelp’s Market Penetration

	High Yelp penetration		Low Yelp penetration	
	Log total admissions (1)	Log Medicare admissions (2)	Log total admissions (3)	Log Medicare admissions (4)
<i>Rated</i>	−0.001 (0.049)	−0.115 ⁺ (0.061)	−0.052 ⁺ (0.027)	−0.100 ^{**} (0.035)
<i>YelpRating</i> × <i>Rated</i>	0.021 (0.016)	0.069 ^{***} (0.020)	0.023 [*] (0.009)	0.033 ^{**} (0.012)
<i>GovRating</i>	0.020 ⁺ (0.011)	0.008 (0.015)	0.014 ^{***} (0.004)	0.022 ^{***} (0.005)
<i>1-star Yelp Rating</i>	0.019	−0.047	−0.028	−0.067 ^{**}
<i>5-star Yelp Rating</i>	0.102 ^{**}	0.228 ^{***}	0.065 ^{**}	0.064 [*]
<i>F-test: YelpRating</i> × <i>Rated-GovRating</i>	0.001	0.060 [*]	0.009	0.011
Time-varying controls	Yes	Yes	Yes	Yes
Facility fixed effects	Yes	Yes	Yes	Yes
Year fixed effects	Yes	Yes	Yes	Yes
Observations	14,040	14,040	103,851	103,851
<i>R</i> ²	0.860	0.827	0.876	0.863

Notes. The table shows the 2SLS results of Equation (1) using IVs for both Yelp and government ratings. Standard errors in parentheses are clustered at the nursing home level.
⁺*p* < 0.1; ^{*}*p* < 0.05; ^{**}*p* < 0.01; ^{***}*p* < 0.001.

obtain the list of all hospitals in the United States from the American Hospital Association database and calculate a hospital’s rating ratio using its cumulative number of consumer ratings on Yelp over its cumulative number of patient admissions. Next, we take the average of these rating ratios over all existing hospitals in a county in 2011 as the measure of Yelp’s market penetration. We then categorize all counties into markets with high and low Yelp penetration using the 75th percentile of their Yelp penetration levels as the cutoff.

We use hospital ratings to measure Yelp’s market penetration for two reasons. First, both hospitals and nursing homes are healthcare facilities, and thus are likely to behave similarly in getting attention from the public on a social media platform. However, Yelp ratings of hospitals would not directly affect nursing homes’ resident admissions. Second, we have a complete list of hospitals in the United States, but such lists cannot be easily obtained for restaurants or hotels. Therefore, using hospital ratings allows us to get a relatively accurate measure of Yelp’s market penetration.¹¹

Table 4 shows the results using two subsamples: markets with high or low Yelp penetration. We focus on the *F* test for *YelpRating* × *Rated* – *GovRating*, that is, the difference between the marginal effect of a one-star increase in the Yelp rating and a one-star increase in the government rating. In markets with low Yelp penetration, the differential effect on Medicare admissions becomes statistically insignificant as shown in column (2), suggesting that consumer awareness of government ratings is as low as that of Yelp in markets where the presence of Yelp is low. By contrast, in markets with high Yelp penetration, column (4) of Table 5 shows that a one-star increase in the Yelp rating increases a nursing

home’s Medicare admissions by 6.0% more than a one-star increase in the government rating. There is no significant differential effect on total admissions regardless of Yelp’s market penetration, as shown in columns (1) and (3). Overall, the results in Table 4 support Hypothesis 3 that the dominance of Yelp ratings over government ratings is stronger in markets with higher Yelp penetration.

7.2. Consumer Literacy

In this section, we explore consumer preference over Yelp and government ratings from the lens of consumer literacy. We conjecture that in markets with lower consumer education level, the influence of Yelp ratings on consumer choice would be stronger, compared with government ratings. There are three reasons for this conjecture. First, consumers with a lower education level might be less likely to know about government ratings. Second, consumers with a higher education level are likely better at processing various quality measures on NHC and thus more receptive to government ratings than those with a low education level (Jin et al. 2022b). Third, well-educated consumers may be more skeptical about Yelp ratings given the limited number of consumer ratings and the existence of fake reviews. For these reasons, we developed Hypothesis 4 in Section 3.3.2 to conjecture that the dominance of Yelp ratings over government ratings is stronger in markets with lower consumer education level.

To proceed, we collect population education information at the county level from the U.S. 2010 Census and use the percentage of residents age 25 and above with a bachelor degree to represent the education level in a county. We then construct two subsamples by categorizing counties

Table 5. Heterogeneous Analysis: Consumer Education Level

	Low educational area		High educational area	
	Log total admissions (1)	Log Medicare admissions (2)	Log total admissions (3)	Log Medicare admissions (4)
<i>Rated</i>	−0.077 (0.048)	−0.186** (0.063)	−0.043+ (0.026)	−0.075* (0.033)
<i>YelpRating</i> × <i>Rated</i>	0.033* (0.017)	0.066** (0.022)	0.022* (0.009)	0.035** (0.012)
<i>GovRating</i>	0.019*** (0.005)	0.020** (0.006)	0.011* (0.005)	0.020** (0.006)
<i>1-star Yelp Rating</i>	−0.044	−0.120**	−0.021	−0.040+
<i>5-star Yelp Rating</i>	0.090*	0.144*	0.066**	0.098***
<i>F-test: YelpRating</i> × <i>Rated</i> − <i>GovRating</i>	0.014	0.046+	0.011	0.015
Time-varying controls	Yes	Yes	Yes	Yes
Facility fixed effects	Yes	Yes	Yes	Yes
Year fixed effects	Yes	Yes	Yes	Yes
Observations	53,037	53,037	64,854	64,854
<i>R</i> ²	0.860	0.847	0.865	0.853

Notes. The table shows the 2SLS results of Equation (1) using IVs for both Yelp and government ratings. Standard errors in parentheses are clustered at the nursing home level.
+*p* < 0.1; **p* < 0.05; ***p* < 0.01; ****p* < 0.001.

into high- and low-education markets using the 75th percentile of their education levels as the cutoff.

Table 5 reports the results of estimating Equation (1) using the two subsamples. Again, we focus on the *F* test of *YelpRating* × *Rated* − *GovRating*. Column (2) shows that in low education markets, a one-star increase in the Yelp rating increases a nursing home’s Medicare admissions by 4.6% more than a one-star increase in the government rating. However, column (4) shows that in high education markets, the differential effect on Medicare admissions between the two ratings becomes statistically insignificant and quantitatively small (1.5%). There is no significant differential effect on total admissions regardless of consumer education level, as shown in columns (1) and (3). To demonstrate the robustness of this result, we categorize counties into high, medium, and low education markets as an alternative market segmentation. We also use the percentage of residents age 65 and above with a bachelor degree as an alternative measure of education level. These results are both qualitatively and quantitatively similar to those in Table 5 (see Table A13 in the online appendix). Specifically, the dominance of Yelp ratings over government ratings was present in markets with both low and medium education levels but not in markets with high education level. Therefore, we find evidence to support Hypothesis 4 that the dominance of Yelp ratings over government ratings is more pronounced in markets with lower consumer education level.

8. Managerial Implications for Nursing Homes

Based on the results in Sections 6 and 7, we aim to derive managerial insights from the perspective of nursing

home operators. Some additional estimations will help us achieve this goal. First, we estimate the impact of Yelp ratings on the financial performance of nursing homes. Second, we analyze what nursing homes have done in response to their Yelp ratings.

8.1. Effect of Yelp Ratings on Financial Performance

To examine the effect of Yelp ratings on nursing homes’ financial performance, we adopt the same regression model as specified by Equation (1) using net revenue and net income as the dependent variable. Specifically, net revenue is defined as the revenue after deducting contractual allowances and discounts from total revenue. Net income is defined as the profit after subtracting all expenses from net revenue. We report the results in Table 6.

Columns (1) and (2) show that increasing a nursing home’s Yelp rating by one star increases its revenue by 1.7%. A five-star Yelp rating increases a nursing home’s revenue by 4.1%, whereas a one-star Yelp rating decreases its revenue by 2.6% compared with those not being rated. These results can be explained by those in Table 2, which shows that a five-star (one-star) Yelp rating increases (decreases) a nursing home’s total admissions and Medicare admissions and hence increases (decreases) its revenue. Columns (3) and (4) show the effect on net incomes. Focusing on the results using IV, we see that a one-star increase in the Yelp rating increases a nursing home’s net income by \$33.5K on average.

8.2. Nursing Homes’ Responses to Yelp Ratings

We have just shown that high Yelp ratings may bring financial benefits to nursing homes. One may wonder

Table 6. Effect of Yelp Ratings on Financial Performance

	Log revenue		Net income	
	OLS (1)	IV (2)	OLS (3)	IV (4)
<i>Rated</i>	−0.021*** (0.006)	−0.043*** (0.011)	−131.306*** (30.455)	−146.479** (54.921)
<i>YelpRating</i> × <i>Rated</i>	0.009*** (0.002)	0.017*** (0.004)	27.806** (9.159)	33.454+ (19.692)
<i>GovRating</i>	0.007*** (0.001)	0.007*** (0.001)	36.729*** (2.649)	36.703*** (2.651)
<i>1-star Yelp Rating</i>	−0.012**	−0.026***	−103.5***	−113.026**
<i>5-star Yelp Rating</i>	0.022***	0.041***	7.723	20.789
Time-varying controls	Yes	Yes	Yes	Yes
Facility fixed effects	Yes	Yes	Yes	Yes
Year fixed effects	Yes	Yes	Yes	Yes
Observations	117,891	117,891	117,891	117,891
R ²	0.953	0.953	0.518	0.518

Notes. The table shows the estimation results of Equation (1). Standard errors in parentheses are clustered at the nursing home level.

+ $p < 0.1$; * $p < 0.05$; ** $p < 0.01$; *** $p < 0.001$.

whether nursing homes made any changes in response to their Yelp ratings. We address this question from two perspectives: online review management and off-line quality improvement.

We randomly sampled 100 rated nursing homes by year 2018 and manually checked whether they responded to consumer reviews on Yelp. We found that only 4 of the 100 nursing homes did. Among those nursing homes whose ratings were below three stars, only 1 of 56 responded on Yelp. These numbers suggest that nursing homes in general pay little attention to their online profiles on Yelp.

To investigate whether nursing homes have improved quality in response to their Yelp ratings, we estimate Equation (1) using the total number of deficiency citations or 30-day rehospitalization rate as the dependent

variable. Both measures have been widely used as quality measures in the nursing home literature (Lu and Lu 2017, 2022). The regression results are reported in Table 7. Overall, nursing homes have not improved quality significantly in response to Yelp ratings. The coefficients of *YelpRating* × *Rated* are negative and insignificant for both quality measures. One possible explanation is that the number of consumer ratings per nursing home on Yelp is relatively low, thereby not attracting enough attention of nursing homes because they probably did not recognize the effect of Yelp ratings on their resident admissions and financial performance could be substantial.

This evidence suggests that nursing homes might not be aware of or have paid little attention to Yelp ratings. Given the potential benefits of improving Yelp ratings as revealed by our study, we recommend that

Table 7. Effect of Yelp Ratings on Quality

	Total citations		30-day rehospitalization	
	OLS (1)	IV (2)	OLS (3)	IV (4)
<i>Rated</i>	0.140 (0.166)	0.172 (0.325)	−0.007*** (0.002)	−0.003 (0.003)
<i>YelpRating</i> × <i>Rated</i>	−0.043 (0.047)	−0.054 (0.111)	0.001+ (0.001)	−0.001 (0.001)
<i>GovRating</i>	0.134*** (0.016)	0.134*** (0.016)	−0.001*** (0.0002)	−0.001*** (0.0002)
Time-varying controls	Yes	Yes	Yes	Yes
Facility fixed effects	Yes	Yes	Yes	Yes
Year fixed effects	Yes	Yes	Yes	Yes
Observations	104,792	104,792	104,792	104,792
R ²	0.509	0.509	0.587	0.587

Notes. The table shows the estimation results of Equation (1). Standard errors in parentheses are clustered at the nursing home level.

+ $p < 0.1$; * $p < 0.05$; ** $p < 0.01$; *** $p < 0.001$.

nursing homes pay close attention to and actively manage their online consumer ratings.

9. Conclusion and Discussion

The goal of this study is to understand how social media and government report cards might differ in their influence on consumer demand of healthcare services. To achieve this goal, we study consumer ratings of nursing homes on Yelp, one of the largest social media platforms, and government ratings of nursing homes on Nursing Home Compare, an online U.S. federal government report card system. We are able to derive actionable insights for both policymakers and nursing home operators.

One of our key findings is that consumer ratings on Yelp exert a stronger effect on nursing home demand than government ratings on Nursing Home Compare. We conjecture that the dominance may be driven by Yelp's brand recognition and consumer literacy. Results from our heterogenous analyses support these two conjectures. This finding is particularly surprising given the fact that only a small number of consumer ratings exist per rated nursing home and only around a quarter of nursing homes have been rated on Yelp, whereas CMS has invested significant resources in establishing Nursing Home Compare with disclose quality information of all U.S. nursing homes to the public. During the COVID-19 pandemic, the public has developed some mistrust of government ratings on Nursing Home Compare (Silver-Greenberg and Gebeloff 2021). Therefore, it is urgent for CMS to take action to restore the public's perception of its five-star ratings. We believe it might be beneficial for CMS to recognize the importance of social media platforms as alternative sources of quality information and to consider collaborating with well-known social media platforms such as Yelp to increase the public awareness of government report card systems such as Nursing Home Compare. Similar efforts have shown success in the restaurant industry, where displaying government hygiene scores on Yelp leads to a significant decrease in consumers' purchase intentions for restaurants with low hygiene scores (Dai and Luca 2020). Furthermore, we recommend that CMS enhance the transparency of nursing home quality reports by incorporating patient satisfaction metrics. For example, the CAHPS nursing home surveys, conducted by the Agency for Healthcare Research and Quality, offer valuable insights into the experiences of nursing home residents and their families. By using survey results, CMS can identify areas that need improvement and track changes in resident satisfaction over time. Moreover, we recommend that CMS consider including comments from previous nursing home residents on their website. These narratives can offer a more personal perspective and supplement the standardized government ratings.

In addition to the policy implications, our study demonstrates the substantial effect of Yelp ratings on nursing homes' financial performance. Although it appears financially beneficial to improve their Yelp ratings, we find little evidence that nursing homes tried to increase their Yelp ratings by actively managing consumer reviews on Yelp or by improving their service quality. Therefore, our study calls for the attention of nursing home operators to recognize the strong influence of social media on consumer demand. Furthermore, we recommend that nursing homes proactively manage their reputation on social media by promptly addressing consumer complaints and implementing quality improvement measures.

Furthermore, our study provides valuable insights into industries where consumers have access to both government and user-generated information channels, which can offer unique quality information to consumers. We highlight a few examples of such industries, including the UK healthcare industry, where the Care Quality Commission regulates, inspects, and rates health and social services, whereas Doctify provides verified patient reviews for doctors, hospitals, and care homes.

Similarly, in the education industry, state governments like New York and Massachusetts disclose school report cards, whereas Niche offers reviews from current students, alumni, and parents. In the airline industry, the U.S. Department of Transportation has published airline on-time performance reports and rankings for decades, while platforms like Skytrax and US News provide reviews and ratings for airlines. We believe it is worthwhile to investigate how different information channels impact consumer choices in these industries, as it can offer valuable insights into the effectiveness of government ratings and the value of user-generated content.

Our study is an initial step toward understanding how consumers choose healthcare services when facing different quality information channels. However, our study has some limitations, which calls for future research on this topic. First, we only observe resident admissions of nursing homes but not their actual demand. Nevertheless, we control for occupancy rates of nursing homes and focus on Medicare admissions, which are generally not subject to capacity constraints. Second, Yelp is not the only online consumer review platform that provides nursing home information to the public. For example, Google also provides consumer reviews for nursing homes. We choose Yelp because it is one of the largest online consumer review platforms. Third, we do not have resident-level data to understand individual consumer behavior in depth. Behavioral experiments can be developed to understand how consumers digest quality information from different channels.

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Endnotes

¹ See <https://www.medicare.gov/care-compare/> (accessed March 20, 2021).

² See <https://www.yelp-press.com/company/fast-facts/default.aspx> (accessed March 20, 2021).

³ See <https://www.reviewtrackers.com/fake-yelp-reviews/> (accessed March 20, 2021).

⁴ See <https://searchengineland.com/yelp-adds-link-to-evidence-of-review-fraud-222740> (accessed March 20, 2021).

⁵ The percentages of hospital-based nursing homes in the OSCAR data and the Yelp data are about 5.8% and 6.0%, respectively.

⁶ The results remain robust when we use the full sample. We also note that the effect of ratings on admissions becomes insignificant when we limit our samples to nursing homes with occupancy rates above the 95th percentile. The tables are available upon request.

⁷ Please see https://www.yelp-support.com/article/What-is-Yelp-s-recommendation-software?l=en_US. Those “unrecommended” reviews have been filtered by Yelp automatically.

⁸ The total effect of a n -star Yelp rating is coefficient of $\text{Rated} + n \times$ coefficient of $\text{YelpRating} \times \text{Rated}$.

⁹ See https://blog.yelp.com/2018/01/yelp_category_list (accessed March 20, 2021).

¹⁰ See <https://www.yelp-press.com/company/fast-facts/default.aspx> (accessed March 20, 2021).

¹¹ It is worth noting that we use year 2011 to construct the penetration measure because it is a relatively early year in our sample period, and it gives us a static measure so that all observations of a nursing home over time belong to the same subsample. We also note that we use the values at a relatively early year because Yelp’s market penetration might be affected by facility marketing behavior as Yelp gradually became known by the public. The underlying assumption of this approach is that Yelp’s penetration in early years is highly correlated with its future penetration. Nevertheless, our results are robust when using an alternative year. Moreover, we utilize the Yelp city list (see <https://www.yelp.com/locations>) to construct an alternative measure of Yelp’s market penetration. We classify a city as a market with high Yelp penetration if the city is listed on the Yelp city list or is located in California (i.e., the location of Yelp’s headquarter). The remaining cities are classified as markets with low Yelp penetration. These results are shown in Table A12 in the online appendix.

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